

Monday July 25, 2022

Duration: 75 minutes

Total: 30 pts

Instructor: Dr. Ali Farid

Student Name:

Student Number:

Signature:

Exercise 1: /3 pts

Exercise 2: /3 pts

Exercise 3: /8 pts

Exercise 4: /6 pts

Exercise 5: /10 pts

Total: /30 pts

Notes:

1. This is a closed book exam with no electronic devices allowed
2. Write your Name and Student Number, and sign this first page
3. Write your name on each of the 3 pages plus the cover page
4. Answer all questions in the space provided or use the back side of the sheet as needed
5. You must submit this examination booklet and sign the attendance sheet
6. You must maintain the confidentiality of your examination; do not provide any opportunity for others to copy any of your work

Exercise 1: Short Questions (3 pts)

Define the following concepts in your own words.

1. Min-heap versus Max-heap trees.
2. D-heap versus Min-Max heap trees.
3. Counting sort versus Quicksort.

Exercise 2: Algorithm Analysis (3 pts)

The function below returns true if there are three equal integers in the array A.

```
int FindTriples(const Vector<int> & A) {  
    for (int i = 0; i < A.Length(); i++)  
        for (int j = i + 1; j < A.Length(); j++)  
            for (int k = j + 1; k < A.Length(); k++)  
                if (A[i] == A[j] && A[i] == A[k] && A[j] == A[k])  
                    return 1;  
    return 0; }
```

1. What is the running time of this function?
2. If it takes ten seconds to run FindTriples on an array of 100 elements, approximately how long will it take to run on an array of 400 elements?

Exercise 3: Binary Tree (8 pts)

Given the binary tree shown in Figure 1, answer the following questions:

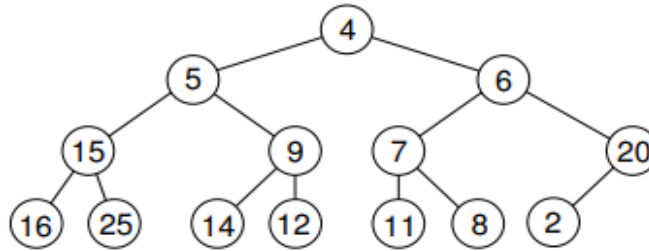
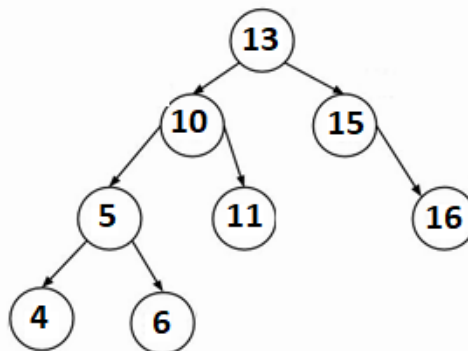


Figure 1: A binary tree T

- Is the tree T a heap? Explain why (not) in at most one (1) paragraph.
- Is the tree T a Binary Search Tree (BST)? Explain why (not) in at most one (1) paragraph.
- Using the notation $(A \leftrightarrow B)$ to represent an exchange show the exchange steps to reheapify the tree. (Note: here the smallest number is the root!)

Exercise 4: AVL tree (6 pts)

Consider the following AVL tree



- Insert 14 into the given tree
- Insert 3 into the given tree
- Insert 7 into the given tree

Show the entire AVL tree after every insertion. You must show complete steps to get full marks.

Exercise 5: Sorting (10 pts)

Answer the following questions I to IV:

I) Which sorting algorithm has the most efficient worst-case runtime?

- (a) Insertion sort
- (b) Selection sort
- (c) Merge sort
- (d) Bubble sort

II) Suppose we are sorting the list [3, 10, 4, 12, 6, 9, 11, 5]. If we use quicksort and the first partition value is 10, which of the following could NOT be the ordering of the list after first partitioning?

- (a) [3, 4, 5, 6, 9, 10, 11, 12]
- (b) [9, 6, 5, 4, 3, 10, 12, 11]
- (c) [3, 4, 5, 6, 9, 11, 10, 12]

III) Which of the following sorting algorithms has the worst case running time of $O(n \log n)$?

- 1) Insertion sort
- 2) Quick sort
- 3) Selection sort
- 4) Merge sort

IV) Show how to apply sorting algorithms below, draw diagrams showing how they sort the given array based on diagrams in class. Sort the array

$$A = \{5, 6, 1, 3, 8, 12, 11, 13\}$$

using:

- a) Quick sort where the pivot is 11
- b) Merge sort

You must show complete steps to get full marks.